



EXAMINATIONS COUNCIL OF ESWATINI
Eswatini General Certificate of Secondary Education

CANDIDATE
NAME

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CENTRE
NUMBER

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CANDIDATE
NUMBER

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PHYSICAL SCIENCE

Paper 3 Practical Test

6888/03

October/November 2025

1 hour 15 minutes

Candidates answer on the Question Paper.

Additional materials: As listed in confidential instructions.
Calculator

READ THESE INSTRUCTIONS FIRST

Write your name, centre number and candidate number in spaces provided.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams, graphs, tables or rough working.
Do **not** use staples, paper clips, highlighters, glue or correction fluid.
Do **not** write on the bar code.

Answer **all** questions.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.
The total number of marks is 40.
Chemistry practical notes for this paper are printed on page 11.

For Examiner's Use	
1	
2	
Total	

This document consists of **11** printed pages and **1** blank page.

- 1** You are going to investigate the reactivity of four metals and their reactions with dilute hydrochloric acid and aqueous silver nitrate by completing two experiments.

You are provided with:

- 1 g each of copper powder, magnesium powder, zinc powder and iron filings
- 1 g each of copper turnings, magnesium ribbon, zinc granules and iron turnings
- 50 cm³ of dilute hydrochloric acid in a beaker
- 40 cm³ of aqueous silver nitrate.

- (a)** In this first experiment, you are going to investigate the reactivity of the metals, copper, magnesium, zinc, and iron with dilute hydrochloric acid.

- Measure the temperature of the dilute hydrochloric acid in the beaker using a thermometer.

- (i)** Record this temperature as the initial temperature in Table 1.1.

[1]

- (ii)** Carry out the following procedure and record your results in Table 1.1.

1. Pour 10 cm³ of the acid into each of four test-tubes labelled **A**, **B**, **C** and **D**, as shown in Fig. 1.1.

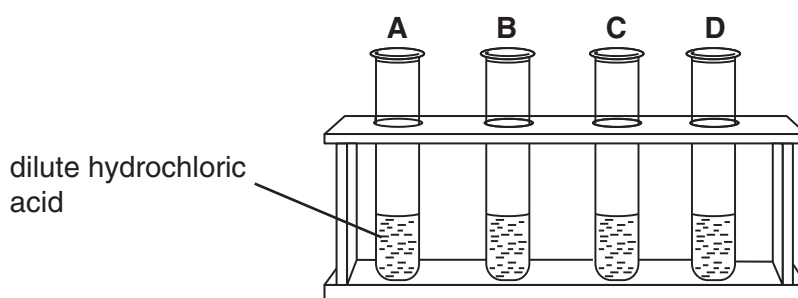


Fig. 1.1

2. Add the iron filings into test-tube **A** and stir gently with the thermometer for three minutes.
3. Read the final temperature and record in Table 1.1.
4. Record one other observation in Table 1.1.
5. Repeat steps **2**, **3** and **4** for test-tube **B** using zinc powder instead of iron filings.
6. Repeat steps **2**, **3** and **4** for test-tube **C** using copper powder instead of iron filings.

7. • Add the magnesium powder into test-tube **D** and stir with the thermometer.
- Insert a lighted splint into the test-tube to test the gas given off and record your observation in Table 1.1.
- Read and record the final temperature in Table 1.1.
- Record **one** other observation in Table 1.1.

Table 1.1

experiment in test-tube	temperature of acid / °C		observations
	initial	final	
A			
B			
C			
D			result with lighted splint other observation

[5]

(iii) Name the gas that is produced in test-tube **D**.

..... [1]

(iv) Using your results in Table 1.1, identify the most reactive metal.

..... [1]

(v) State **two** reasons for your answer to (a)(iv).

1

2

[2]

(vi) Identify a variable that was controlled to ensure the experiment was a fair test.

..... [1]

- (b) In this second experiment you are going to investigate the reactivity of the metals copper, magnesium, zinc, and iron with aqueous silver nitrate.
1. Pour 7 cm³ aqueous silver nitrate into each of four test-tubes labelled **E**, **F**, **G** and **H**. Measure the temperature of the silver nitrate in each test-tube using a thermometer and record this in Table 1.2.
 2. Add zinc granules to test-tube **E**, iron turnings to test-tube **F**, copper turnings to test-tube **G** and magnesium ribbon to test-tube **H** simultaneously.
 3. Leave the test-tubes undisturbed for **three minutes**.
 4. Measure the temperature of the contents of the test-tubes.
- (i) Record the temperature and observations in Table 1.2.

The observation made with zinc has been done for you.

Table 1.2

metal	temperature / °C		observation
	initial	final	
zinc			grey precipitate formed slowly
iron			
copper			
magnesium			

[3]

- (ii) Using your results in Table 1.2, state and explain the type of reaction that occurs.

.....

 [2]

- (iii) Zinc nitrate is one of the substances formed when zinc granules react with aqueous silver nitrate.

Name the other substance formed in the reaction.

..... [1]

- (iv) Using your results from Table 1.2, state which metal is the least reactive.

Explain your answer.

least reactive metal

explanation

..... [2]

- (c) Using the results from (a) and (b), arrange the four metals in order of their reactivity, starting with the most reactive.

most reactive

.....

.....

least reactive

[1]

[Total:20]

- 2 In this experiment you are going to investigate the relationship between a load and the extension of a spring.

Fig. 2.1 shows the set-up of the experiment.

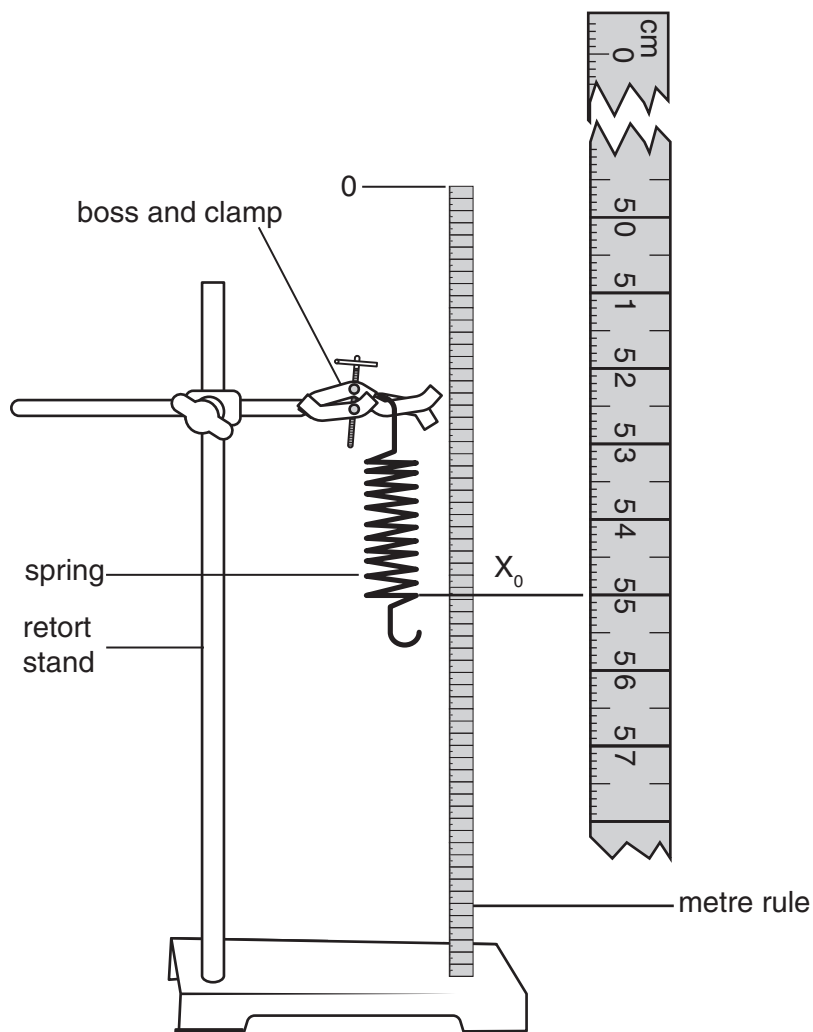


Fig. 2.1

- (a) Read and record the position of the bottom of the spring, X_0 , shown in Fig. 2.1.

[Write your answer in millimetres.]

..... [1]

(b) Suspend the provided 100 g mass on the spring as shown in Fig. 2.2.

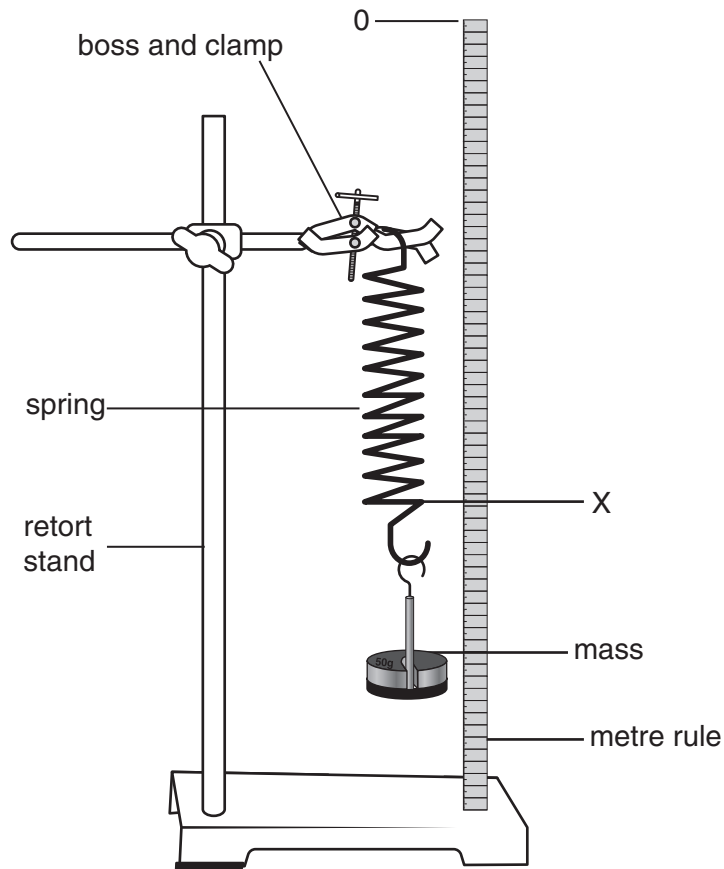


Fig. 2.2

Read and record the new position of the bottom of the spring, X, in Table 2.1.

[1]

- (c) • Repeat procedure (b) by adding 50 g masses, one at a time, on the suspended mass.
- Record the new positions of the bottom of the spring, X, in Table 2.1 for each 50 g mass added.

Table 2.1

mass / kg	0.10	0.15	0.20	0.25	0.30
load / N					
new position 'X' / mm					
extension					

Complete Table 2.1 by:

- (i) filling in the unit for extension [1]

- (ii) calculating the load, W, for each mass using the formula:

$$W = mg$$

[Take $g = 10 \text{ N / kg}$]

[2]

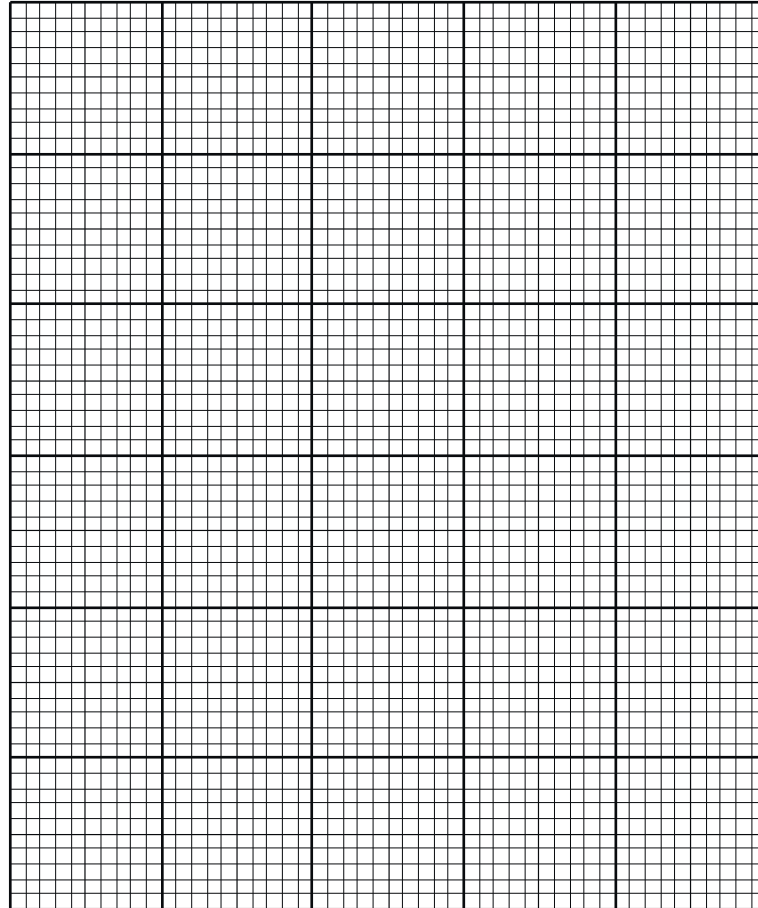
- (iii) calculating the extension, e, caused by each mass using the formula:

$$e = X - X_0.$$

[2]

- (d) Using the results in Table 2.1, plot a graph of extension against load.

extension /



load / N

[3]

- (e) (i) Using the graph, determine the load when the extension is 50 mm.

..... N [2]

- (ii) Using your value in (e)(i) and the extension of 50 mm, calculate the spring constant, k .

Use the formula:

$$W = ke$$

Where: W = load

k = spring constant

e = extension

..... N / m [2]

- (f) State the relationship between the load and the extension of the spring.

.....
..... [1]

- (g) State **one** source of error and the precaution you took to correct it when carrying out the experiment.

source of error
precaution [2]

- (h) Suggest how the method described in (b) and (c) can be changed to investigate the limit of proportionality.

.....
.....
.....
..... [3]

[Total: 20]

CHEMISTRY PRACTICAL NOTES

Tests for anions

<i>Anion</i>	<i>Test</i>	<i>Test result</i>
carbonate (CO_3^{2-})	add dilute acid	effervescence, carbon dioxide produced
chloride (Cl^-) [in solution]	acidify with dilute nitric acid, then add aqueous silver nitrate	white ppt.
iodide (I^-) [in solution]	acidify with dilute nitric acid, and then add aqueous lead(II) nitrate / aqueous silver nitrate	yellow ppt.
nitrate (NO_3^-) [in solution]	add aqueous sodium hydroxide then aluminium foil; warm carefully	ammonia produced
sulfate (SO_4^{2-}) [in solution]	acidify with dilute nitric acid, then add aqueous barium nitrate	white ppt.

Tests for aqueous cations

<i>Cation</i>	<i>Effect of aqueous sodium hydroxide</i>	<i>Effect of aqueous ammonia</i>
ammonium (NH_4^+)	ammonia produced on warming	no reaction
calcium (Ca^{2+})	white ppt., insoluble in excess	no ppt or very slightly white ppt.
copper(II) (Cu^{2+})	light blue ppt., insoluble in excess	light blue ppt., soluble in excess giving a dark blue solution
iron(II) (Fe^{2+})	green ppt., insoluble in excess	green ppt., insoluble in excess
iron(III) (Fe^{3+})	red-brown ppt., insoluble in excess	red-brown ppt., insoluble in excess
zinc (Zn^{2+})	white ppt., soluble in excess giving a colourless solution	white ppt., soluble in excess, giving a colourless solution

Tests for gases

<i>gas</i>	<i>test and test results</i>
ammonia (NH_3)	turns damp litmus paper blue
carbon dioxide (CO_2)	turns limewater milky or white
chlorine (Cl_2)	bleaches damp litmus paper
hydrogen (H_2)	'pops' with a lighted splint
oxygen (O_2)	relights a glowing splint

